# Project Visualization Report

## Final Visualization Development, Recreation, and Comparison

This report explains the completion of the final recreation for the **“World’s Biggest Data Breaches & Hacks”** visualization. It focuses specifically on the development steps, the final recreated visualization, and the comparison between my recreation and the original visualization.

## G. Multiple Steps of the Visualization Process

The visualization was completed in multiple stages inside the Jupyter notebook. The notebook first introduces the project with the title, student information, course name, course number, and instructor name. It also references the original visualization with both a link and an image, then loads all required packages and explains the purpose of any package used for the project.

The first development step was loading the public dataset connected to the original visualization. After the data was loaded, the dataset was cleaned so that the important columns could be used correctly. These columns included organization name, year, records lost, sector, breach method, display value, story/details, and source links. Cleaning was necessary because some values were missing, some column names needed standard formatting, and numeric values such as records lost had to be converted into a usable format for visualization.

After the data cleaning stage, an initial version of the visualization was created to test the basic idea of representing each data breach as a circle. This first step helped confirm that each organization could be displayed as a bubble and that the size of the bubble could represent the number of records lost.

The next step added size scaling. In this stage, larger data breaches were shown as larger bubbles, while smaller breaches were shown as smaller bubbles. This was an important part of the recreation because bubble size is one of the main features of the original visualization.

After size scaling, color encoding was added. Each bubble was colored based on the breach method, such as hacked, poor security, oops, lost device, inside job, or unknown. This helped recreate the visual meaning of the original chart and made the categories easier to compare.

The layout was then improved by arranging the bubbles using a packed bubble style layout. The bubbles were grouped around year labels so the final result looked closer to the original visualization. Instead of a simple scatter plot or bar chart, the visualization uses a more organic bubble arrangement similar to the original Information is Beautiful design.

In later stages, interactivity was added. Hovering over a bubble expands it and shows more information, including the organization name, records lost, and a short description. If a source link is available, the bubble shows “click to read more,” and clicking the bubble opens the source in a new browser tab. I also added a rule so that only one expanded bubble can be open at a time, which makes the interaction cleaner and easier to use.

Additional controls were added to make the visualization more interactive. These include filtering by sector, breach method, year range, organization search, and the number of records displayed. The notebook shows these changes step by step, highlighting how the visualization developed from a simple bubble chart into a more complete interactive recreation.

## H. Final Recreation of the Original Visualization

The final visualization recreates many important features of the original **Information is Beautiful** visualization. Each bubble represents a major data breach or hack, and the size of each bubble is based on the number of records lost. This matches the main design idea of the original visualization, where larger circles represent larger data breaches.

The final version includes many of the largest and most recognizable breaches from the dataset. Large breaches such as National Public Data, Indonesian SIM cards, LinkedIn, Facebook, Ticketmaster, and other major events appear as larger bubbles. Smaller events appear as smaller circles around them. This makes the visualization useful because the viewer can quickly identify which breaches had the largest impact.

The recreated visualization also uses colors to represent breach methods. This follows the original design, where color is used to separate different types of breaches. The legend explains the meaning of each color so the viewer can understand the categories.

The year labels are placed on the side of the chart, and the bubbles are arranged around those year markers. This recreates the original structure where breaches are visually organized by time. However, my recreation makes the year organization clearer than the original because the bubbles are more visibly connected to the year labels on the left side. In the original visualization, the bubbles are packed very tightly, which gives it a strong visual impact, but it can also make it harder to clearly identify which exact year some bubbles belong to. My version keeps the original packed-bubble idea while making the year-based grouping easier to understand.

The final design also includes a header, subtitle, update note, legend, search box, filter controls, and an original-style top gradient border. These visual elements help the recreation look closer to the original presentation. I was able to recreate approximately 90% of the original view, including the main layout, title style, bubble sizing, color categories, legend, and interactive behavior. The remaining 10% includes design choices and improvements that are my own, mainly related to clearer filtering, clearer year separation, and a slightly less compressed bubble layout.

The final visualization includes interactive exploration. When the viewer hovers over a bubble, the selected bubble expands and displays more details inside it. The text includes the organization name, approximate number of records lost, and a short description. If a source link is available, the user can click the bubble to open the original source in a new tab. This feature supports exploration and makes the visualization more useful than a static chart.

The visualization also includes extra filtering and sorting features. Users can filter the bubbles by sector, breach method, year range, organization search, and the number of top records displayed. The sector filter allows users to focus on one industry, such as gaming, healthcare, finance, or technology. The method filter allows users to view only specific types of breaches. The search box allows users to find an organization directly. The Top N control allows users to sort and limit the results based on the number of records lost, so the largest breaches can be viewed more clearly.

One important addition in my recreation is the year range filter. This feature is not present in the same way on the original website. The year filter allows the viewer to focus on a specific time period, such as 2010 to 2025, instead of viewing all years at once. This makes the visualization more useful for analysis because users can compare breaches by time period and reduce visual clutter when needed.

Several layout improvements were also made in the final version. The visualization width was compacted to reduce unnecessary side space. The chart height was made dynamic so that bubbles do not overflow outside the white visualization area. The hover behavior was also improved so only one bubble detail view is open at a time. These changes helped make the final output cleaner and easier to use.

Overall, the final recreation attempts to match the original visualization as closely as possible in terms of bubble size, color, layout, labels, filtering, hover interaction, source-link interaction, and overall visual style. At the same time, it includes a few intentional improvements that make the visualization clearer and more interactive for analysis.

## I. Final Comparison: Errors, Omissions, and Differences from the Original

The final recreation is close to the original visualization, but there are still some differences. Based on the comparison between the original screenshot and my recreated version, I would estimate that my visualization recreates about 90% of the original look and behavior. The remaining 10% represents differences caused by layout limitations and intentional improvements added in my version. The original **World's Biggest Data Breaches & Hacks** visualization appears to use a carefully refined custom layout. My recreation uses a D3 force simulation to approximate the packed bubble arrangement. Because of this, the exact position of every bubble is not identical to the original.

Some bubbles may appear slightly higher, lower, or farther apart compared with the original. The original visualization packs the bubbles more tightly, which creates a dense and polished appearance. My version is slightly less compact because I wanted the bubbles to remain readable, stay inside the white chart area, and connect more clearly with the year labels. This means the overall structure is similar, but the exact bubble arrangement is not a perfect copy.

Another difference is the label placement. In the original, labels appear very carefully fitted inside the bubbles. In my version, labels are generated dynamically based on the size of each bubble. This works well for larger and medium-sized bubbles, but very small bubbles may have limited text or may only show details when hovered over because there is not enough space inside the circle.

The hover interaction is also not exactly the same as the original. My version expands the selected bubble and shows the details inside the bubble. The original may use a different tooltip or interaction style depending on the interactive version being viewed. However, the main purpose of the interaction is preserved because users can explore details, view record counts, read descriptions, and open source links.

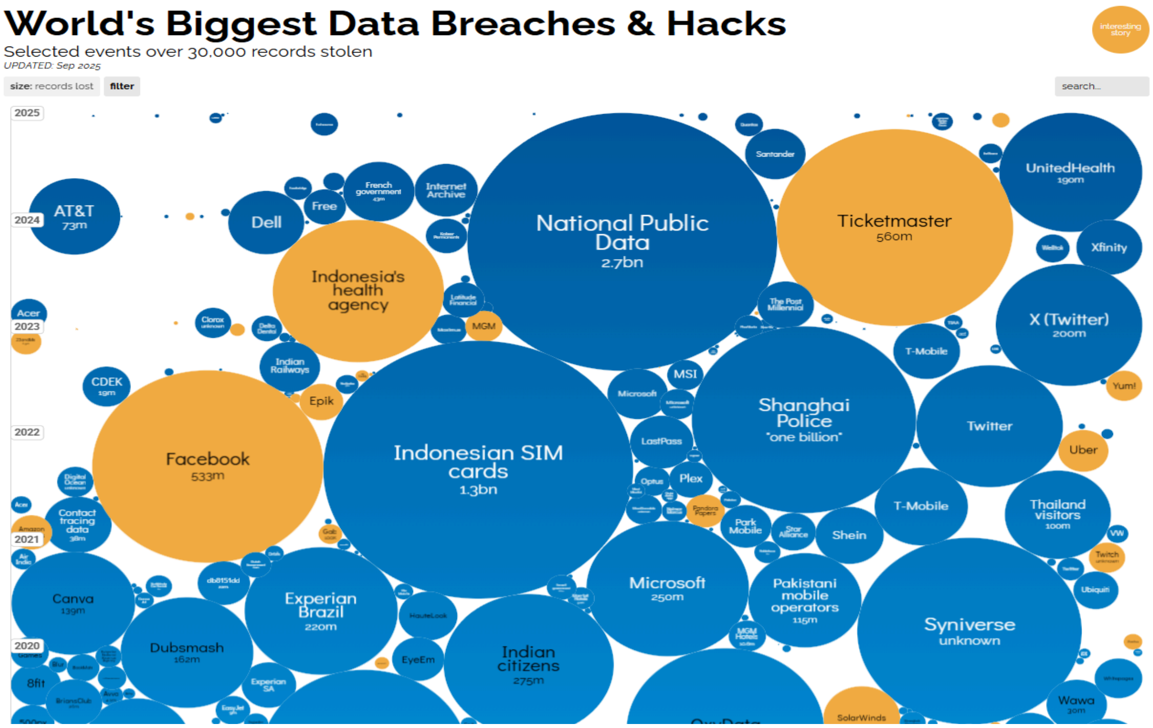
There may also be small differences in typography, spacing, and exact color tones. I attempted to recreate the original title style, subtitle, legend, year labels, top gradient border, and bubble colors, but the browser-rendered result may not be identical to the original image.

One major improvement in my version is the filter system. The original visualization has filtering options, but my recreation adds a clearer filter panel with sector, method, from-year, to-year, Top N, and search controls. The year filter is an additional analytical feature that helps users isolate a specific time range. This makes my version slightly different from the original, but it improves usability and makes the visualization easier to explore.

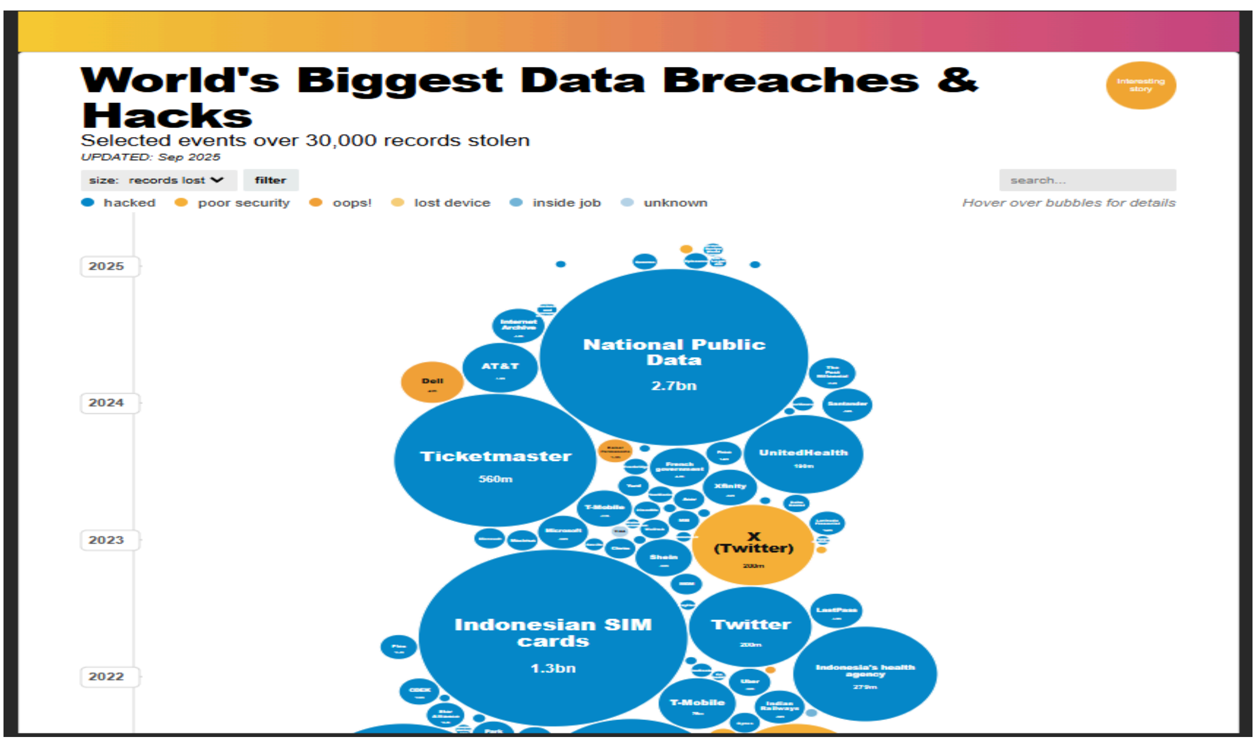
Another difference is the year readability. In the original visualization, all bubbles are very compact and visually impressive, but the dense packing can make it difficult to clearly determine which year some bubbles belong to. In my version, the year labels and bubble positions are easier to follow, so the viewer can better understand the timeline relationship.

Even with these differences, the final recreation includes the major features of the original visualization. It uses a packed bubble layout, year-based organization, size encoding for records lost, color encoding for breach method, labels, filtering, search, hover details, clickable source links, and a final comparison section.

All required project components were completed in the Jupyter notebook. The notebook includes the project title, student name, course name, course number, instructor name, original visualization reference with link and image, package loading, explanation of why the visualization is excellent, data reference and loading, data cleaning with comments, multiple visualization development steps, the final interactive visualization, and this final comparison explaining differences from the original.




This is the original output




This is the notebook output with the improvements

## Completion Statement

This project successfully completes the required notebook sections and final visualization requirements. The final notebook documents the full process from data loading and cleaning to multiple visualization development stages and the final interactive recreation. The final report also explains the strengths of the recreation and honestly identifies the remaining differences between my version and the original visualization.